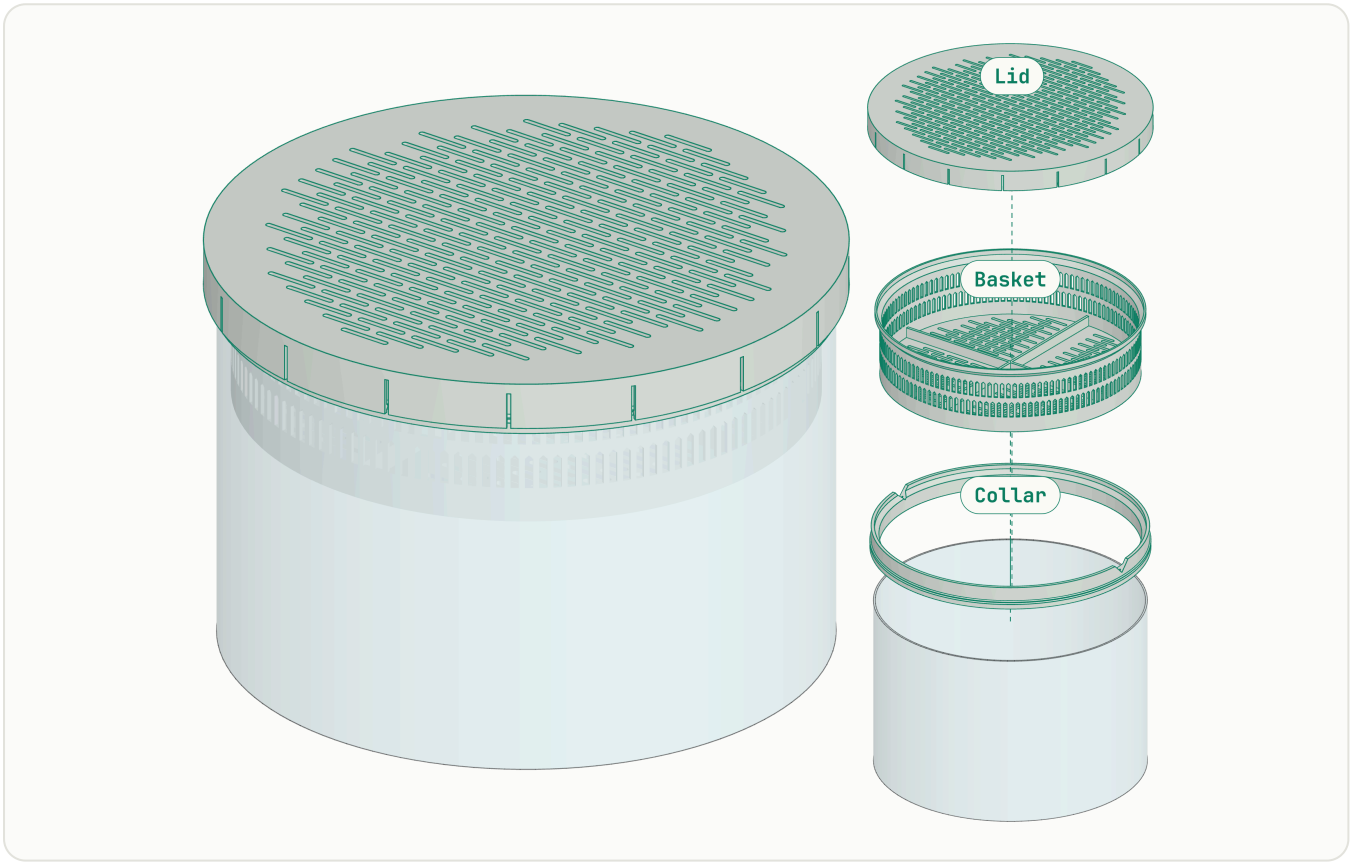
PINK STUFF

A desiccant basket for the bottom half of a pink antifreeze jug

Cut a West Marine -50°F antifreeze gallon jug in half with the cutting ring. A collar goes over the cut edge. A basket of desiccant pellets hangs in the collar. A lid snaps onto the collar and keeps the pellets in. The pellets take water from the air. The brine drips through the floor of the basket and collects in the jug, below the basket.

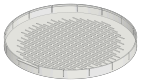


Left: the assembly on the jug. Right: the parts. From the bottom: the jug, the collar, the basket, the lid.

Measured and estimated. The model uses a 152.8 mm diameter, from a 480 mm circumference on a sample jug. The 1.0 mm wall is an estimate. Print the fit test arc first. `MEASURE.md` lists the measurements to take.

- ☐ White PLA
- 80 g and 10 h 20 min for each jug
- cutting ring 45 g, once
- No supports
- No hardware

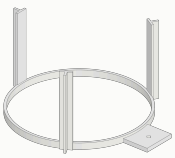
Printed parts

	File	Colour	Size on the bed (mm)	Filament	Print time	Cost
	`stl/jig.stl`	White	200 × 160 × 96	36 g	3 h 30 min	$0.71
	`stl/jig_cart.stl`	White	13 × 46 × 26	9 g	47 min	$0.18
	`stl/jig_pin.stl`	White	8 × 8 × 17	0 g	6 min	$0.01
	`stl/fit_arc.stl`	White	9 × 50 × 13	2 g	15 min	$0.04
	`stl/collar.stl` × 6	White	157 × 157 × 13	115 g	10 h 02 min	$2.30
	`stl/basket.stl` × 6	White	151 × 151 × 40	196 g	30 h 36 min	$3.92
	`stl/lid.stl` × 6	White	159 × 162 × 12	170 g	21 h 26 min	$3.41
	Total			**529 g**	**66 h 40 min**	**$10.58**

Filament and print time from the Printdeck slice: the Printdeck slicer with the TAZ Workhorse SE profile (pla-0.20-standard, white PLA), the same slice as printdeck.beetlebug.org. Cost at $19.99/kg. The counts are for 6 jugs.

Print

The STL files are in their print orientation. Load them, do not turn them.



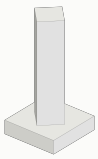
jig

support check 0 g
Upside down, the ring top on the bed. Print it once for all the jugs.



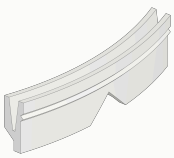
jig cart

support check 0 g
On its back end, the open end of the blade slot on the bed.



jig pin

support check 0 g
Cap on the bed.



fit arc

support check 0 g
Top face on the bed. Print this first. It tests the groove and the diameter on the jug.



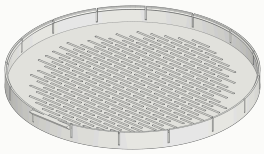
collar

support check 0 g
Top face on the bed (upside down). The groove opens to the top.



basket

support check 0 g
Floor on the bed. The rim flares out at 45°.

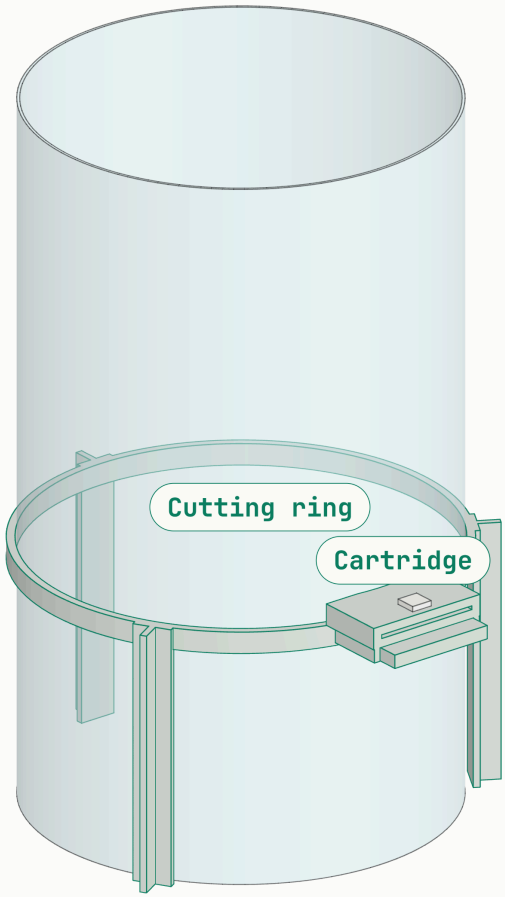


lid

support check 0 g
Top face on the bed (upside down). The skirt goes up.

1 Cut the jug

cutting ring | cartridge | pin | razor blade | jug | flat table

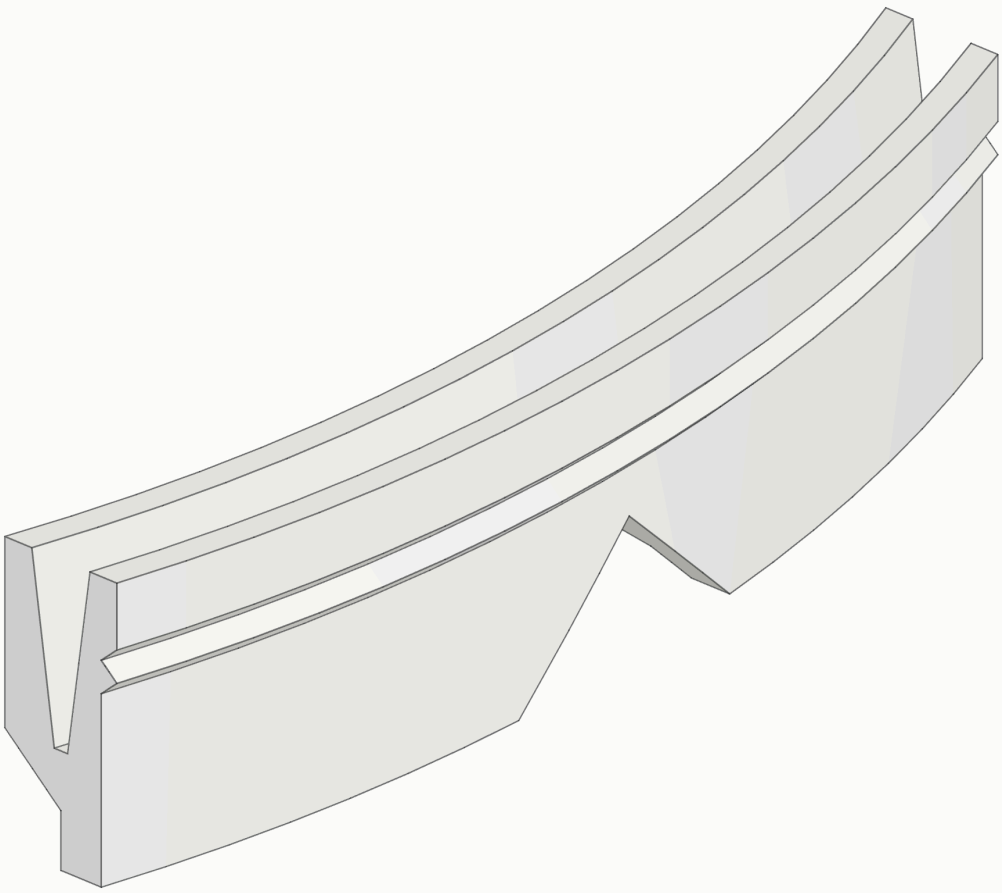


1. Empty the jug. Rinse it with water and let it dry.
2. Stand the cutting ring on a flat table. Put the jug down into the ring from above.
3. Slide the razor blade into the back end of the cartridge, edge first, spine up to its stop. The thick spine goes in only one way.
4. Put the cartridge on the shelf, behind the ring. Push its back end with your thumb. The blade goes through the wall.
5. Drop the pin into the hole. The pin locks the blade and the cartridge.
6. Hold the shelf. Turn the jug by its handle, one full turn, in either direction. The blade cuts a level line.
7. Pull the pin. Pull the cartridge back. Lift off the top half, then lift the jug out.
8. Sand the cut edge flat and remove the burrs. The cut is 100 mm above the table.

Note. The ring centres the jug, so each cut is level and at the same height. The edge goes 2.3 mm into the jug wall (1.7 mm at the least). The edge points into the ring, at the jug, and your thumb pushes behind the blade. Take the blade out of the cartridge after use.

2 Fit test

`fit test arc` `cut jug`

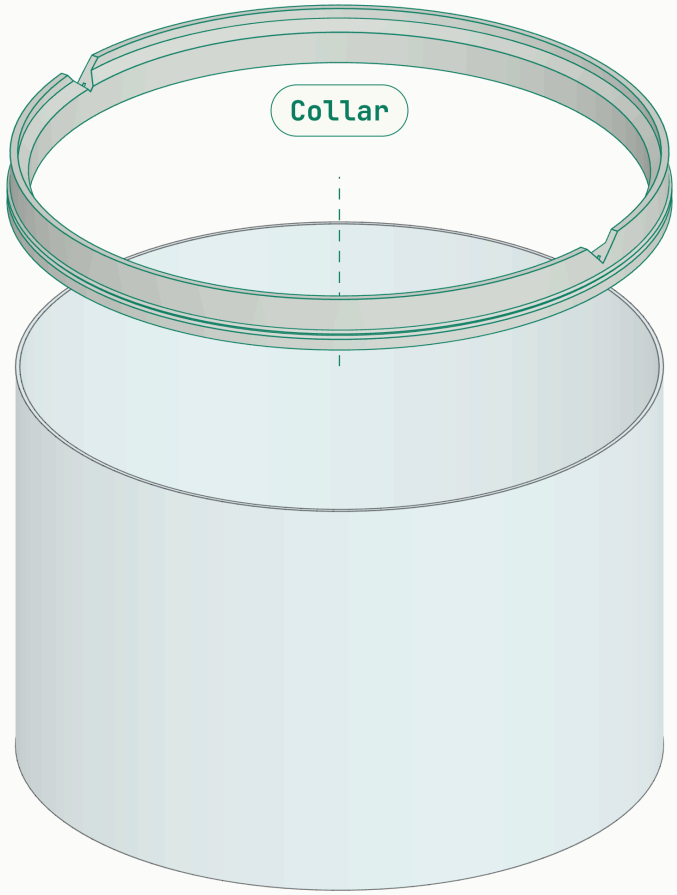


1. Push the groove of the fit test arc down over the cut edge.
2. The edge must go into the groove and stop with a firm push.
3. Look at the outer face. The skirt must follow the jug wall, with no gap at the ends of the arc.

Note. The jug diameter comes from one sample jug. Other jugs can differ. A gap at the ends: the diameter in the model is too large. A gap at the middle: it is too small. Measure the jug (`MEASURE.md`), change `JUG_DIA` in `shape.py`, run `make.sh`, and print the arc again.

3 Put on the collar

collar jug

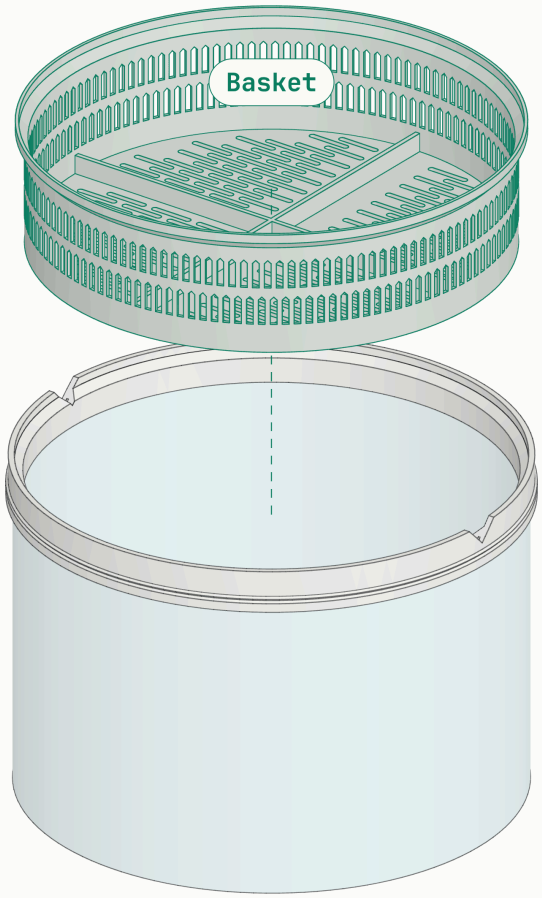


1. Turn the collar so that the groove opens down.
2. Put the groove over the cut edge of the jug. Start at one side and work around.
3. Push down hard all around. The edge wedges into the groove.
4. Pull up on the collar. It must stay on the jug.

Note. The groove tapers from 2.6 mm at its mouth to 0.6 mm at its roof. A wall from 0.6 to 2.4 mm wedges in. The lid snap also presses the collar skirt onto the jug.

4 Put in the basket

`basket` `desiccant pellets`



1. Lower the basket into the collar. Its flared rim sits on the 45° seat of the collar.
2. Fill the basket with desiccant pellets. It holds about 0.6 L. Fill it to 10 mm below the rim.
3. Use calcium chloride pellets, for example DampRid refills. The pellets must be larger than the 2 mm slots.
4. To lift the basket out, put your fingers into the two notches in the collar and pull up on the rim.

Note. The pellets take water from the air and turn to brine. The brine drips through the slots into the jug.

5 Snap on the lid

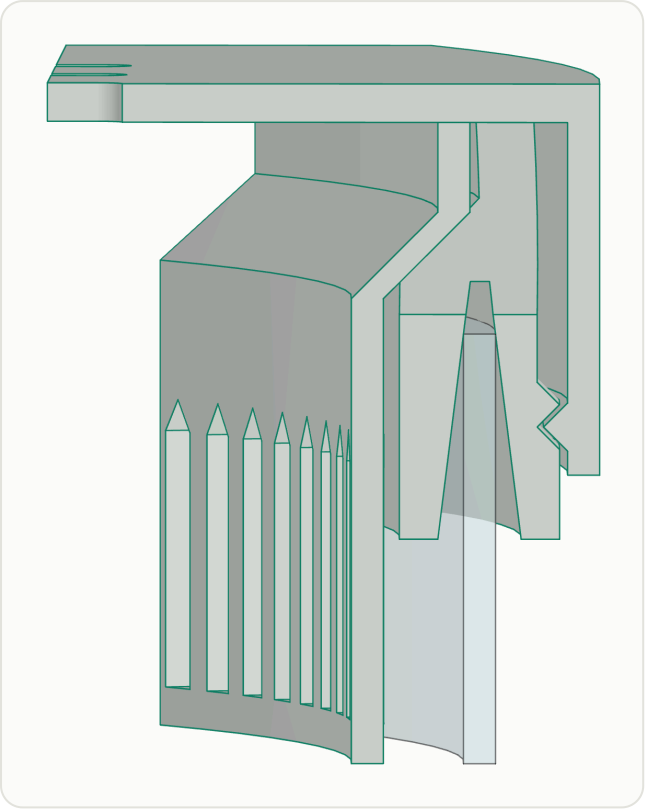
`lid`



1. Put the lid on the collar, with the pull tab at one side.
2. Push down on each side of the lid until it clicks. The bead of the skirt goes over the ridge of the collar.
3. To remove the lid, pull the tab out and up.
4. Empty the brine when it gets near the basket floor. Remove the collar with the basket and the lid. Pour the brine down a drain with water.

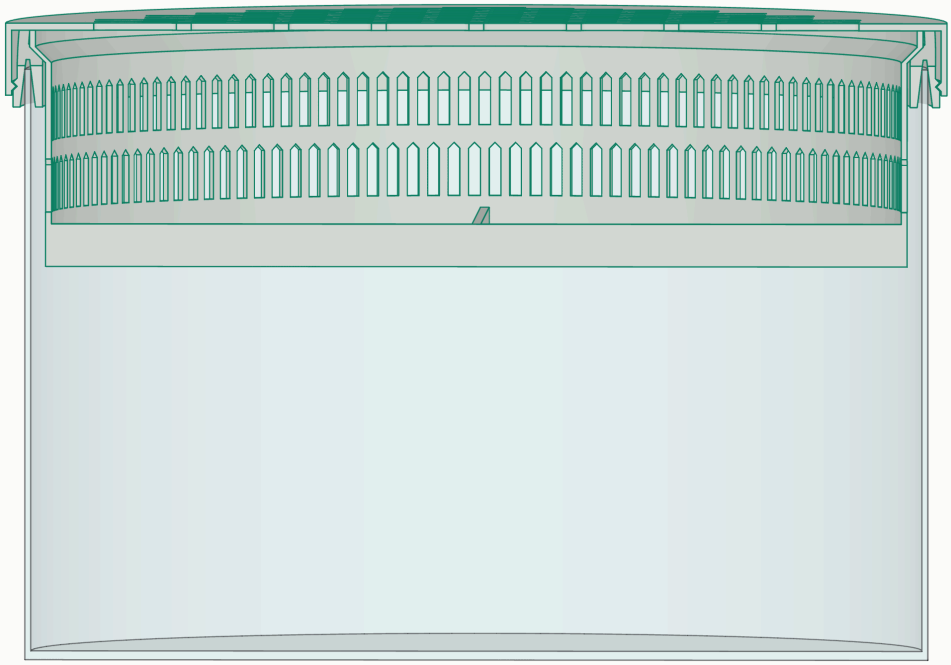
Note. The air slots in the lid are 1.5 mm wide. Air goes in, and the pellets stay in if the jug falls over. Brine can come out if the jug falls over: put the jug in a place where it can not fall.

How it locks



The section through the rim, at one side of the jug. From the left: the basket rim under the lid plate, the collar, the jug wall in the groove, and the lid skirt.

- **Groove.** The cut edge goes up into a tapered groove in the collar. A 1.0 mm wall stops 1.6 mm below the roof. A firm push wedges it in.
- **Seat.** The basket rim flares out at 45°. It sits on a 45° seat in the collar and centres itself.
- **Snap.** A bead inside the lid skirt goes over a ridge on the collar skirt, and clicks into a groove. The snap is 0.5 mm. 16 slits cut the skirt into fingers. Each finger bends with 1.17 % strain (PLA takes about 1.5 %).
- **Clamp.** The lid skirt holds the collar skirt against the jug wall.
- **Rim.** The lid plate sits on the basket rim. The pellets stay in.



The section through the middle. The basket holds 0.6 L of pellets. The space below the basket holds 1.1 L of brine (cut at 100 mm). Two ribs stiffen the floor. The floor has 158 slots and the wall has 260 slots, all 2 mm wide. The wall slots let air reach the pellets and save filament. Their tops are pointed at 45°, so they print with no support.

Adjust the fit

The jug wall is an estimate. Measure the jug (`MEASURE.md`), change the values in `shape.py`, run `make.sh`, and print the fit test arc again.

Problem	Value	Now (mm)	Change
The collar is loose on the jug, or it does not go over the edge	`JUG_DIA`	153	Measure the circumference of the jug at the cut, and divide by 3.1416 (MEASURE.md, J1).
The blade does not cut through the wall	`BLADE_OUT`	2.9	Add 0.5 to BLADE_OUT.
The jug does not go into the cutting ring, or it is loose in it	`RING_CLR`, `JUG_DIA`	0.6, 152.8	Check JUG_DIA (MEASURE.md, J1). Change RING_CLR by 0.2.
The blade is loose in the slot, or it does not go in	`BLADE_T`, `SPINE_T`, `SLOT_CLR`	0.25, 0.9, 0.35	Measure the blade (MEASURE.md, B1 and B2). Add or subtract 0.05 from SLOT_CLR.
The edge stops at the mouth of the groove	`JUG_T`, `G_HALF_MOUTH`	1.0, 1.3	Measure the wall (J2). Set JUG_T to the value. For a wall over 2.2 mm, add 0.3 to G_HALF_MOUTH.
The collar is loose on a correct jug size	`G_HALF_ROOF`	0.3	Subtract 0.1 from G_HALF_ROOF. The wall then wedges higher in the groove.
The lid is too hard to put on, or it is loose	`SNAP`	0.50	Subtract or add 0.1. The skirt strain must stay below 1.5 % (model.py prints it).
The basket sits too high or too low	`DEPTH, CUT_H`	35, 100	Change DEPTH for a deeper or shallower basket. CUT_H is only for the brine volume and the images.

The fit check. `model.py` checks the collar on the jug, the basket in the collar, and the lid on the collar, also on the path in. It stops if a part touches where it must not, if the collar has no grip, or if the lid has no snap.